Topic 11 – Information handling, modelling and planning

11.1 Marketing, innovation and feasibility

Spec: Marketing – marketing analysis, research techniques, raw data/analysed data to enable enterprise to be encouraged; Innovation management – cooperation between management, designers and production engineers, the encouragement of creativity; The use of feasibility studies on the practicability of proposed solutions

Key facts

• **Four parts of a marketing analysis?** Industry overview, target market, competition, pricing and forecast.
• **Digital research techniques?** Online/app surveys, SMS polls, social media analytics, GPS/location-based surveys, website analytics, biometrics (heart rate, eye tracking), virtual environments, video interviews, QR codes.
• **Raw vs analysed data?** Raw = unprocessed responses; analysed = processed into trends, statistics and charts to inform decisions and encourage enterprise.
• **Tools to encourage creativity?** Brainstorming/ideation, design workshops, prototyping (CAD, 3D printing), suggestion schemes, kaizen, cross-functional teams, staff rotation, rewarding ideas, no blame for failure.
• **Why must management, designers and engineers cooperate?** Ideas must be commercially viable (management), desirable and usable (designers) and manufacturable at the right cost and quality (engineers) – avoids costly late redesigns.
• **What does a feasibility study assess?** Technical feasibility (materials, processes), economic (direct/indirect costs, pricing, profit, break-even), market demand, legal/IP/standards, environmental, scheduling and resources.
• **Direct vs indirect costs?** Direct: materials and labour for specific production. Indirect (overheads): rent, lighting, heating, maintenance, admin, auxiliary staff.
• **Market pull vs technology push example?** Pull: demand for longer phone battery life. Push: invention of touchscreens or 3D printing finding new products.

Key terms

Marketing analysis	Part of a business plan/feasibility study describing the industry, target market, competition, and pricing and sales forecast.
Primary research	New data collected first-hand – surveys, interviews, focus groups, observation, user trials.
Secondary research	Existing data collected by others – market reports, government statistics, websites, sales data.
Raw data	Unprocessed information as collected (e.g. individual survey answers).
Analysed data	Data that has been processed, summarised and interpreted (charts, averages, trends) to support decisions.
Quantitative / qualitative data	Quantitative: numerical, measurable (how many). Qualitative: opinions and descriptions (why).
Innovation management	Systematically managing the generation, development and implementation of new ideas, products and processes to improve competitiveness.
Market pull / technology push	Pull: innovation driven by consumer demand. Push: innovation driven by new technology looking for an application.
Feasibility study	An assessment of whether a proposed product/project is practical and viable – technically, financially, in the market, legally and environmentally – before major investment.

Exam tip (Hodder): When answering on market research, explain how **digital technology** gives access to target data from particular demographic groups.

Exam tip: Typical mistake: Don't only discuss whether consumers will like it – cover costs, materials, manufacture and marketing too.

11.2 Modelling the costing of projects

Spec: Budgets – undertake financial forecasts; Planning for production – allocation of employees, materials, scale of production; Selection of appropriate tools, machines and manufacturing processes

Key facts

• **Break-even formula?** Break-even quantity = fixed costs ÷ (selling price − variable cost per unit).
• **Considerations when preparing budgets?** Set-up, materials, labour, transport, training, manufacturing, overheads, marketing, production capacity, market trends, break-even, price, quality, demand, storage, packaging, energy, inflation, profit targets, government policy, contingency.
• **What are financial forecasts based on?** Past and current income/expenditure, market trends, government policies, long-term aims, sales/inventory/manufacturing reviews.
• **Planning for production – what is allocated?** Employees (numbers, skills, shifts), materials (quantities, stock, JIT ordering, waste allowance) and scale of production (one-off/batch/mass).
• **Why choose die casting rather than sand casting for a high-volume product?** Faster cycle, better accuracy and finish, thinner walls, less machining; high die cost spread over large volume.
• **Why are mass-market car bodies not made with rivets and adhesive?** Steel panels can be robot spot welded quickly and cheaply; mechanical fasteners and adhesives are slower and more labour intensive (used for aluminium premium cars).
• **Cost of an aluminium bracket: what stages?** Material cost (volume × density × price) + waste % + tooling + labour (hours × rate) = net cost; then add profit %.

Key terms

Budget	A financial plan setting out expected income and expenditure over a period.
Financial forecast	A prediction of future sales, costs, cash flow and profit, based on past data, market trends and plans.
Fixed costs	Costs that do not change with output – rent, tooling, machinery, salaries, insurance.
Variable costs	Costs that rise with output – materials, components, packaging, energy per unit, piecework labour.
Break-even point	The output at which total revenue equals total costs – no profit or loss. Break-even = fixed costs ÷ (price − variable cost per unit).
Profit	Revenue − total costs. Profit margin = profit ÷ revenue × 100.
Mark-up	Percentage added to cost to set the selling price.
Cash flow	The movement of money in and out of a business over time.

Exam tip: Costing calculations (2021 Q4c – 8 marks) reward each correct **stage**. Show every step with units, and keep unrounded values until the end.

11.3 Intellectual property rights

Spec: The importance, implications and ways of protecting the intellectual property rights of designers, inventors and companies:; Patents, copyrights, design rights, trademarks

Key facts

• **Which IPR protects how a product works?** A patent.
• **Which IPR protects appearance?** Registered design (and unregistered design right).
• **Which IPR is automatic and protects drawings, photos and software?** Copyright (and unregistered design right for shape).
• **Forms a trademark may take?** Logo/symbol, name/initials, word/slogan, sound/jingle, colour scheme, shape, device.
• **Disadvantages of patenting?** Expensive and slow (3–4 years), details become public, competitors work around it, must defend in court, limited to 20 years and country of application, may be outdated by the time granted.
• **Ways copyright protects a company's images?** Prevents others copying, distributing, selling, lending, showing publicly, putting online, or adapting them; supports legal action.
• **How can designers protect the form and function of a product?** Patent (function) and registered design/design right (form). Not copyright or trademark.
• **Durations?** Patent 20 yrs; registered design 25 yrs (5-year renewals); unregistered design right up to 15 yrs; copyright life + 70 yrs; trademark 10 yrs renewable indefinitely.

	Patent	Copyright	Registered design	Unregistered design right	Trademark
Protects	How it **works** – technical/functional features of an invention	Original written, artistic, musical, photographic, digital works – drawings, images, software, manuals, websites	**Appearance** – shape, contours, pattern, colour, texture, or namentation	Shape and configuration (3D form) of a product	**Brand identity** – logo, name, slogan, sound, colour, shape

	Patent	Copyright	Registered design	Unregistered design right	Trademark
How obtained	Apply to IPO (detailed drawings and description, often via a patent attorney); must be new, inventive, industrially applicable	**Automatic** on creation (use ©, dates, watermarks, keep drafts as evidence)	Register with IPO (illustrations + fee)	**Automatic**	Register with IPO (fee)
Duration	Up to **20 years** (annual renewal fees)	Creator's life + **70 years**	Up to **25 years** (renew every 5)	Up to 10 years after first sale (max 15 years)	**10 years**, renewable indefinitely
Symbol	"Patent pending"/patent number	©	Registered design number	–	™ / ®
Example	Dyson cyclone, kettle thermostat, phone features	Product photos, CAD drawings, instruction booklets, website	Shape of a Coca-Cola bottle, a chair design	Shape of a new bottle opener	Nike swoosh, "Just Do It", Intel jingle, Cadbury purple

Key terms

Intellectual property (IP)	Creations of the mind – inventions, designs, artistic works, brand names – that can be owned and protected by law.
Patent	Protects how a new invention works (its function/technical features) for up to 20 years if renewed. Must be new, inventive and capable of industrial application. Applied for through the IPO.
Copyright	Automatic protection for original written, artistic, musical, photographic and digital works (including drawings, software, websites) – usually for the creator's life + 70 years. ©
Registered design	Protects the appearance (shape, pattern, colour, ornamentation) of a product for up to 25 years, renewed every 5 years, by registering with the IPO.
Unregistered design right	Automatic UK protection for the shape/configuration of a product against copying – up to 10 years after first sale (15 years after creation). Harder to prove.
Trademark	Registered sign that distinguishes a company's goods – logo, name, slogan, colour, sound, shape. Renewed every 10 years indefinitely. ™ unregistered, ® registered.
IPO	UK Intellectual Property Office – grants patents, registered designs and trademarks.
Licensing	Allowing another company to use your IP in exchange for payment (royalties).

Exam tip: Use one product to show all four rights – e.g. a pen: **patent** on the retractable mechanism, **registered design** for its shape, **copyright** on the packaging graphics and website, **trademark** on the brand name and logo. In 2023 Q4c, only **patents and design rights/registered designs** protect form and function – copyright and trademarks were rejected.

11.4 Standards

Spec: Implication to designers, manufacturers and consumers of the following standards when developing designs and manufacturing products:; British Standards (BSI and Kitemark); European (CEN and CE); International Standards (ISO)

Key facts

• **What does the Kitemark show?** The product has been independently tested by BSI and meets the relevant British Standard; the manufacturer's processes are regularly audited.
• **What does the CE mark mean?** The manufacturer declares the product conforms to EU health, safety and environmental requirements; mandatory for EU sale of many products.
• **Name three standards organisations UK manufacturers follow.** BSI, CEN, ISO (also ANSI for US markets).
• **Examples of British Standards?** BS 1363 (13 A plugs and sockets), BS EN 71 (toy safety), BS 8888 (technical drawing), BS 7176 (furniture fire resistance), BS EN 1078 (cycle helmets).
• **Examples of ISO standards?** ISO 9001 quality management, ISO 14001 environmental management, ISO 50001 energy management, ISO 7250 anthropometric measurements, ISO 216 paper sizes (A4).
• **Implications of standards for designers?** Must research and design to relevant standards from the start (dimensions, materials, safety features, testing), can constrain creativity, but give clear guidance and credibility.
• **Implications for manufacturers?** Testing and certification costs, audits and documentation, but access to UK/EU/global markets, fewer recalls, legal compliance, consumer trust and competitive advantage.
• **Implications for consumers?** Confidence in safety and quality, compatibility/interchangeability of parts (plugs, batteries, paper sizes), easier comparison; may slightly raise prices.

Designers	Manufacturers	Consumers
Must identify and design to relevant standards from the start (dimensions, materials, safety features, labelling)	Testing, certification and audit costs; documentation; possible redesign	Confidence that products are safe and reliable
Standards may constrain creativity but provide proven guidance and data (e.g. anthropometrics)	Access to UK, EU and world markets; legal compliance	Compatibility and interchangeability of parts
Products gain credibility and are easier to market	Fewer recalls, claims and fines; protects reputation; competitive advantage	Easier to compare products; marks such as Kitemark/CE/Lion Mark guide purchases
Must keep up to date as standards change	Need ongoing QC to maintain certification	Slightly higher prices to cover compliance

Key terms

Standard	An agreed document setting out specifications, test methods, dimensions or procedures that products, services or systems should meet.
BSI	British Standards Institution – the UK national standards body; publishes British Standards (BS) and awards the Kitemark.
Kitemark	BSI certification mark showing a product has been independently tested and the manufacturer's quality systems audited against a British Standard.
CEN	European Committee for Standardisation – produces European Standards (EN). British Standards adopted from them carry the prefix BS EN.
CE mark	Conformité Européenne – manufacturer's declaration that a product meets EU safety, health and environmental requirements; required to sell many products in the EU/EEA.
UKCA mark	UK Conformity Assessed – the post-Brexit UK equivalent of the CE mark for Great Britain (CE is still accepted for many goods).
ISO	International Organization for Standardization – global body of national standards bodies (including BSI) producing international standards, e.g. ISO 9001 (quality), ISO 14001 (environment).

Exam tip: Typical mistake: Vague points about "safety" score little – use a specific standard (e.g. BS 1363 plugs) and explain its features.